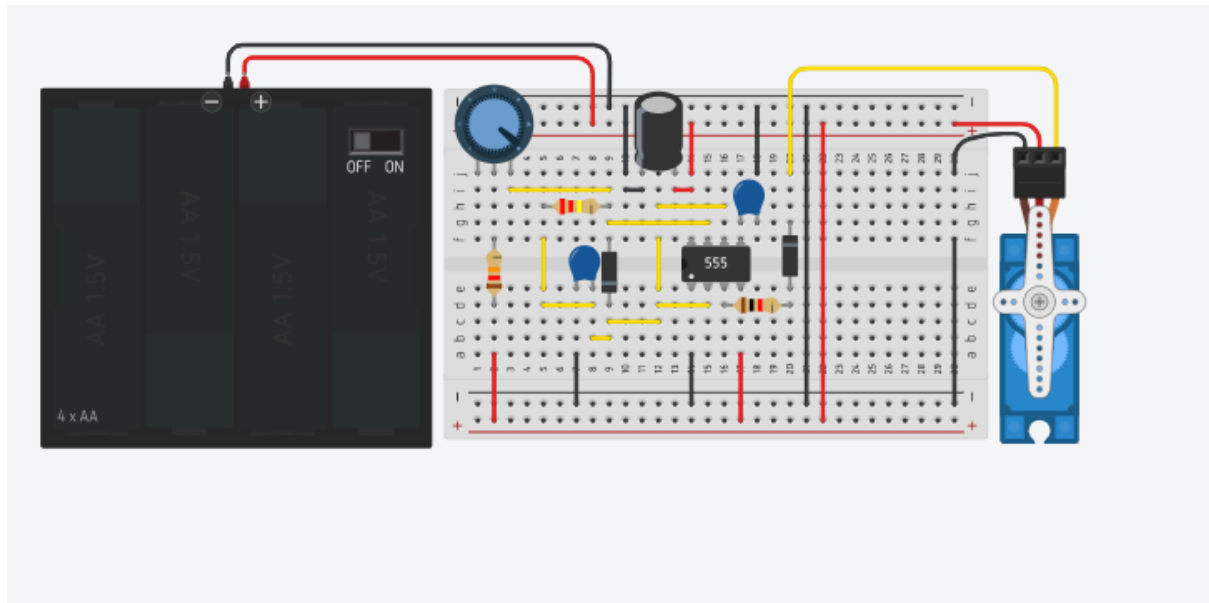


# ***Pulse Width Modulation Control using opamp and 555 timer***

## **ABSTRACT :**

1. INTRODUCTION
2. ABOUT OP AMP -- 555 TIMER--SERVO WORKING
3. WHAT IS PWM?
4. POTENTIOMETER → 555 TIMER --> SERVO
5. SIMULATION CIRCUIT, COMPONENTS USED AND WHY?; SCHEMATICS.
6. PROBLEM STATEMENT AND OUR APPORACH.

## TINKERCAD SIMULATION CIRCUIT



## COMPONENTS USED

| Component | Value         | Purpose              | Why Chosen                           |
|-----------|---------------|----------------------|--------------------------------------|
| R3        | 220k $\Omega$ | Oscillation resistor | Sets frequency with C2               |
| R1        | 1k $\Omega$   | Discharge control    | Works with R3 & C2                   |
| C2        | 100nF         | Timing capacitor     | Sets frequency                       |
| C3        | 100nF         | Noise filter         | Standard on Pin 5                    |
| R2        | 12k $\Omega$  | Pulse width bias     | Works with POT                       |
| RPOT1     | 20k $\Omega$  | Pulse width adjust   | Fine tuning                          |
| D1/D2     | -             | Protection diodes    | Reverse voltage & noise protection   |
| C1        | 10 $\mu$ F    | Power filter cap     | Smooths voltage for stable operation |
| BAT1      | 6V            | Power source         | Standard for servo & 555             |
| Servo1    | -             | Output actuator      | Controlled by PWM from 555           |

**SCHEMATIC DIAGRAM :**

